Pulkit Girdhar

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Education

B.Tech. **Indian Institute of Technology Roorkee**, Electrical Engineering — Roorkee, India, 2024 – 2028
- CGPA: 9.00/10 (expected graduation 2028)
- Relevant coursework: Numerical Methods, Probability Methods, Data Science, Signals & Systems

Class XII **B.G.N. Adarsh Public School, Bathinda**, Class XII (CBSE) — Bathinda, India, 2024
- 90.60%

Class X **St. Xavier's School, Bathinda**, Class X (CBSE) — Bathinda, India, 2022
- 92.00%

Experience

Trust Lab, IIT Bombay, Research Intern, advised by Prof. Pratik Jawanpuria (C-MInDS) — May 2026 – July 2026, 3 months
Context compression and adaptation in prior-data fitted networks (TabPFN).
- Built a transformer attention-analysis pipeline that extracts TabPFN's internal attention distributions to rank training samples by informativeness for compressed inference.
- Fine-tuned LoRA adapters on a frozen TabPFN across synthetic priors from four data-generating families; cross-prior transfer degraded as the Wasserstein/MMD distance between priors grew.
- Related prior geometry to adapter geometry with adapter-similarity metrics, tested this under cross-family ablations, and caught and corrected an undertraining artifact that inflated the effect.

Department of Electrical Engineering, IIT Roorkee, Undergraduate Researcher, advised by Prof. Parikshit Pareek — Roorkee, India, Feb 2026 – Apr 2026, 2 months
Gaussian-process surrogates for EV-charging feasibility on power grids.
- Replaced repeated AC power-flow solves with GPU-accelerated Gaussian-process surrogates (BoTorch, CUDA) for rapid EV-charging feasibility screening.
- Generated AC power-flow datasets in Julia across loading conditions; surrogates reached about 1e-5 p.u. voltage error and 1e-4 rad angle error.
- Used GP predictive uncertainty (squared-exponential kernel) for voltage-constraint screening, delivered through an interactive feasibility-analysis tool.

Publications

LoRA Adaptation of Tabular Foundation Models for AC Power Flow
Pulkit Girdhar, Pratik Jawanpuria, Parikshit Pareek.
Pulkit Girdhar, Pratik Jawanpuria, Parikshit Pareek
(Submitted to the AAAI-27 Student Abstract Program (under review))

Projects

Variational Quantum Classifiers — Dec 2025 – Jan 2026
Quantum Computing Group, IIT Roorkee. PennyLane variational quantum classifiers benchmarked against classical baselines.
- Data re-uploading ansatz with circular CNOT entanglement, class-weighted loss, early stopping on validation AUC, stratified splits and train-only scaling.
- Benchmarked against logistic regression and random forest on credit-card fraud and breast-cancer data.

Hydrogen Orbital Solver Dec 2024 – Feb 2025

Advised by Prof. Abhiraj Sharma, Civil Engineering, IIT Roorkee.

- Solved hydrogen-atom wavefunctions from the Schrödinger equation with Legendre, associated Legendre and Laguerre recursions implemented from scratch.
- 3D orbital visualizations across quantum numbers (n, l, m).

Positions of Responsibility

- **Executive Member**, Quantum Computing Group, IIT Roorkee (Feb 2025 – present): organise technical sessions on quantum computing, help run Qiskit and PennyLane workshops, and mentor junior members.
- **Member**, FuMES Special Interest Group: lead sessions and write handouts.
- **Member, Technical Council**, Vivekananda Bhawan, IIT Roorkee (Nov 2025 – present).

Skills

Languages

ML & scientific computing

Research tooling

Interests

Research interests